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CellVoyager: AI CompBio agent generates new insights by autonomously analyzing biological data

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Modern biology increasingly relies on complex, high-dimensional datasets such as single-cell RNA sequencing (scRNA-seq), which present a vast space of potential hypotheses. Systematically exploring this space is often impractical, as scRNA-seq analyses are time-consuming and require substantial computational and domain expertise. To address this challenge, we introduce CellVoyager, an AI agent built on large language models that autonomously generates and implements scRNA-seq analyses within a Jupyter notebook environment. We evaluate CellVoyager on CellBench, a benchmark of 76 published scRNA-seq studies, where it outperforms GPT-4o and o3-mini by up to 23% in predicting which analyses authors ultimately conducted, given only the papers' background sections. Across three in-depth case studies, CellVoyager generated novel findings in COVID-19, cell–cell communication and aging that experts consistently rated as creative and scientifically sound. These results demonstrate CellVoyager's potential to accelerate computational biology and uncover missing insights by autonomously analyzing biological data at scale.

Modern biological research often involves collecting and analyzing complex, high-dimensional datasets. These datasets range from multiomics assays to spatial and temporal profiling, each harboring rich biological insights[1–3]. Yet, the high dimensionality that enriches these datasets also introduces substantial analytical challenges, requiring complex, domain-specific computational methods to extract real biological signal[2,4–6]. Many of these methods have steep learning curves, especially for researchers without computational backgrounds. Furthermore, there are numerous biological hypotheses that can be formalized using the various combinations of features in these high-dimensional datasets. Many of these hypotheses may remain unexplored because of the vast scale of this space of analyses and the expertise necessary to execute some of these analyses.

Recent advances in large language models (LLMs) have demonstrated strong capabilities in both biological reasoning[7,8] and complex code generation[9]. These developments have spurred the emergence of LLM-based artificial intelligence (AI) agents as a new paradigm for scientific data analysis, where natural language instructions are translated into executable code[10–12]. Beyond executing user-specified tasks, AI agents have demonstrated potential in helping researchers navigate the complex landscape of possible analyses by generating hypotheses[13–16] and by effectively using diverse analytical and computational methods[17–19].

However, in a typical biological research setting, investigators have a set of analyses they plan to try execute or have already executed. Therefore, a truly collaborative AI agent should not only respond to user prompts in isolation but should also complement the investigator's prior efforts. Although recent LLM-based agents have shown promise in scientific hypothesis generation or automated analysis execution, to our knowledge, they have not been shown to reliably

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and autonomously generate and test new analytical hypotheses that meaningfully build upon a researcher's prior analyses.

Single-cell RNA sequencing (scRNA-seq) provides an ideal testbed for studying these challenges. By measuring gene expression at a single-cell resolution, scRNA-seq has enabled the identification of pathogenic cell subpopulations, inference of cell state transitions and discovery of differentially expressed genes across biological conditions[20–22]. However, the high dimensionality and presence of technical and biological confounders necessitate substantial computational expertise for rigorous analysis[23]. This complexity has led to the development of thousands of open-source computational tools for analyzing scRNA-seq data[24]. Moreover, the extensive gene coverage and high resolution of scRNA-seq data enables the formulation and exploration of a large set of biological hypotheses. Collectively, these result in a vast space of possible computational analyses that can be conducted on the data. Depending on the time, resources and expertise available to a set of researchers, they may be limited to a narrow subset of these analytical workflows, potentially missing important biological findings. AI agents capable of autonomous dataset exploration could help address these issues by leveraging the extensive biological and computational knowledge of their underlying LLMs to explore a wide range of analyses. This autonomous exploration could also help find biological insights that are counterintuitive and that a researcher may not think to test.

In this Article, we introduce CellVoyager: an LLM-driven AI agent that autonomously explores scRNA-seq datasets. By taking in the processed scRNA-seq dataset and a record of analyses already performed on the dataset, CellVoyager generates and executes new analytical pipelines that build directly on prior work. Unlike earlier systems, we focus on how well CellVoyager can complement and extend an existing set of analyses, facilitating a more collaborative human-AI research workflow. This work focuses on using published research papers as input into CellVoyager, as this represents a challenging setting where extensive effort has been put into exploring the dataset. To demonstrate the utility of CellVoyager, we applied it to a set of published scRNA-seq studies, where several PhD-level researchers evaluated the agent's findings, including some of the original authors of the respective manuscripts.

Results

CellVoyager methodology

CellVoyager is an agentic framework that autonomously explores scRNA-seq datasets while incorporating the user's past analyses to prevent redundancy and help find analyses the user may be unfamiliar with. By integrating an LLM with live code execution within a Jupyter environment, CellVoyager dynamically generates, iteratively refines and executes novel analysis plans-termed 'exploration blueprints'. In this work, CellVoyager is based on OpenAI's o3-mini model, chosen for its strong performance on scientific coding benchmarks and relatively low computational cost[25]. A schematic of the CellVoyager agentic workflow is provided in Fig. 1, while a more detailed description of the agent, including the prompts used, is provided in the Methods.

Agent setup. CellVoyager requires two primary inputs: a processed scRNA-seq dataset and a report that explains the biological context of the dataset, including any prior analyses. The report contextualizes the core biological question, summarizes what is in the single-cell dataset and describes the prior computational single-cell analyses performed on the dataset. In this work, we use a published manuscript as the report because of the challenge in finding unexplored analytical directions in them. CellVoyager can also optionally invoke OpenAI's deep researcher to retrieve and synthesize relevant biological background information. This capability allows the agent to incorporate domain-specific knowledge and identify previously performed analyses in the literature, helping it to prioritize novel or complementary analyses.

To guarantee reproducibility and prevent dependency conflicts, CellVoyager operates within a fixed Jupyter kernel[26] that includes popular single-cell packages part of the scverse[27] such as scanpy and scvi-tools[28,29], as well as more general packages such as seaborn[30]. The agent first generates a summary of the paper, separating it into (1) biological background; (2) analyses attempted; and (3) details about the dataset used. Moving forward, CellVoyager uses this summary rather than the full text of the paper. The agent also generates a Jupyter notebook, initializing the execution environment by loading in the dataset and the relevant Python packages. By interfacing with an interactive notebook, CellVoyager is able to access previous results, data and intermediate outputs, allowing it to efficiently iterate on its analysis plan.

Analysis planning, execution and replanning. CellVoyager operates through creating and reflecting upon 'exploration blueprints'. Each blueprint has three components: (1) a hypothesis; (2) a step-wise analysis plan; and (3) the Python code required for the next step.

Before execution, CellVoyager first self-critiques the blueprint, identifying potential weaknesses in the plan and retrieving documentation from the Python functions referenced in the code to verify that they are being used appropriately. The blueprint and code are then revised as needed by the agent, after which the code is appended to the Jupyter notebook for execution. If the code execution fails, the agent iteratively rewrites the code for up to F attempts; if errors persist, CellVoyager is told to revise its analytical trajectory. In this work, we set $F = 3$ as, empirically, the percentage of code cells successfully fixed saturates after that mark (Supplementary Fig. 1). If the code runs successfully, CellVoyager interprets the images and text outputted by the code through a vision language model, which in this work is GPT-4o. From its interpretation of these results, the agent then replans the next steps in its current exploration blueprint as needed. For each execution output, the agent also provides a natural language interpretation of the results and appends it as a markdown block to the notebook. A visualization of the process is provided in Fig. 1a,b.

In the Jupyter notebook, at each step, CellVoyager's basic analysis block consists of a description of its current action, a code cell executing that action, the outputs from running the code cell and an interpretation of the outputs. An example of a CellVoyager-generated analysis is shown in Fig. 1c,d, which includes the agent's interpretation of the figure.

CellVoyager runs analyses sequentially, where each analysis is conditioned on the previous analyses to ensure that CellVoyager does not repeat analyses. After all analyses are complete, the agent extracts the most promising results and writes a report detailing its hypotheses, procedure and interpretation of those results. This allows the agent to effectively explore many analyses and then highlight a subset that are scientifically interesting to the user. As the agent's full exploration trajectory is documented in the Jupyter notebook, with headers detailing the agent's hypothesis and results interpretation at every iteration, the notebook acts as a detailed and interpretable full report if the user desires more detail.

Evaluating CellVoyager analysis generation with CellBench

To evaluate the ability of CellVoyager to propose relevant analyses for a research topic, we introduce CellBench: a benchmark for measuring how well an agent can infer analyses in a single-cell transcriptomics research paper given only the paper's background section. Each query in CellBench is derived from a published single-cell study, where we use an off-the-shelf LLM (gemini-2.5-pro-preview) to extract the biological background from the paper and to generate a list of all of the single-cell computational analyses reported in the paper. The biological background section acts as the input context for each CellBench query. Given this context, the agent is tasked to propose single-cell analyses that it thinks the paper should perform. The analyses actually performed in this paper act as the ground truth analyses. The overlap

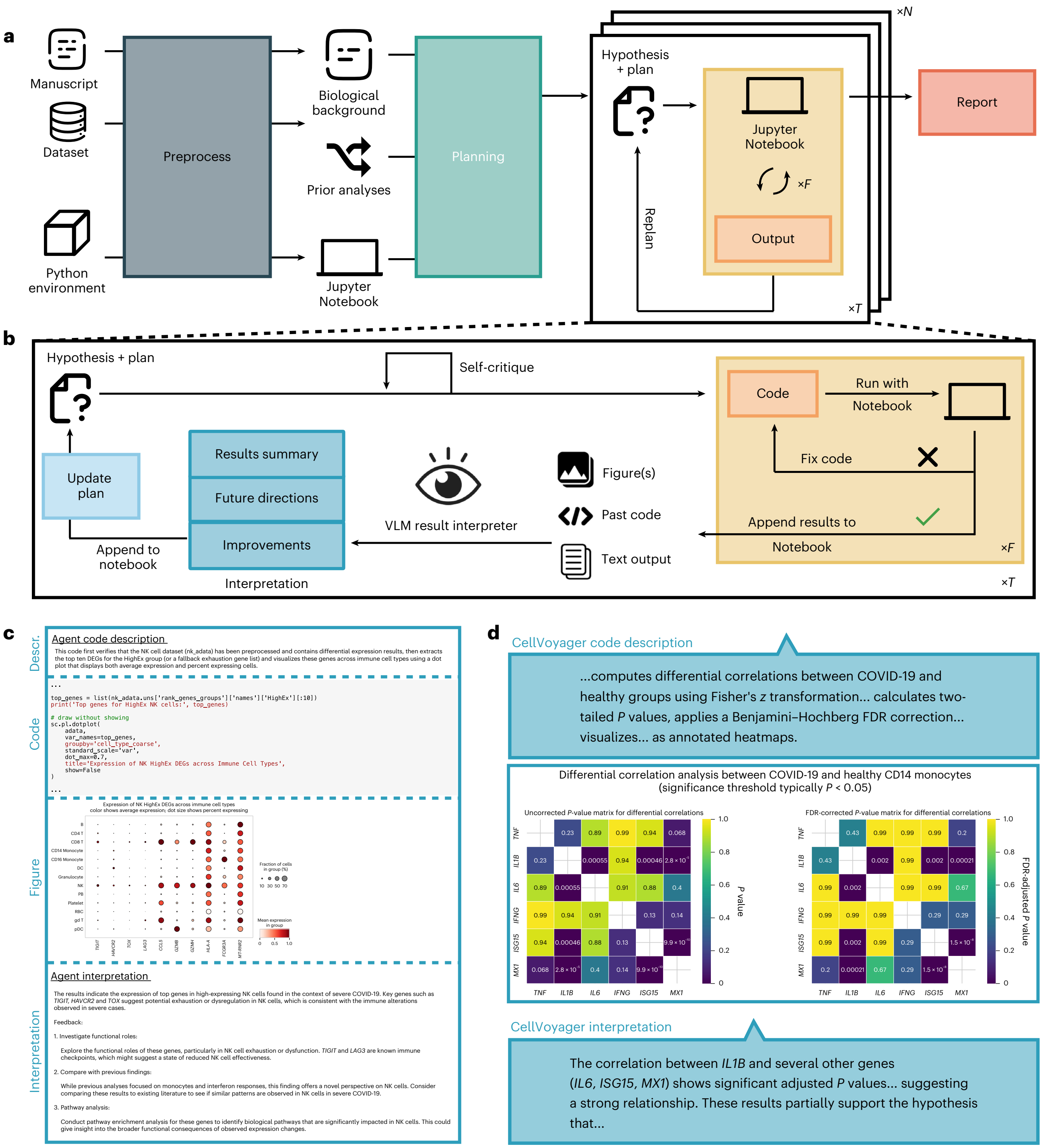

Fig. 1 | Schematic of the CellVoyager agentic framework. a, Overall schematic of CellVoyager. (1) The manuscript text, associated scRNA-seq data and a Python environment specification are given as input. The biological background and prior analyses performed in the manuscript are extracted and a Jupyter notebook is initialized. (2) Exploration blueprints (consisting of a hypothesis and a stepwise plan) are generated. The agent self-critiques this blueprint, incorporating any suggested improvements. (3) Each hypothesis is explored for *T* steps. Each step involves generating and executing code, with up to *F* steps of bug fixing, replanning on the basis of the code execution output and self-critiquing the new plan. This process can be repeated for *N* distinct hypotheses, with the agent being given past hypotheses it tested to prevent it from repeating the same analyses. A final report is generated that summarizes noteworthy findings from the *N* exploration trajectories. **b**, Overview of CellVoyager analysis and interpretation module. The self-critiqued plan contains Python code for the next step of the analysis. This code is executed and the outputs from running the code, along with the contents of past code cells, are fed into a vision language model (VLM) for interpretation. The VLM outputs a summary of the outputs, suggesting future directions on the basis of promising results and possible ways to improve the analysis; these inform how the agent decides to choose the next step in the analysis and CellVoyager updates its analysis plan accordingly. **c**, Basic analysis block of CellVoyager in a Jupyter notebook, consisting of a description of the analysis step, the code to carry out the analysis, the outputs and figures and the interpretation of the outputs and figures. NK, natural killer; DEG, differentially expressed gene. **d**, An example of an analysis block of CellVoyager, with code omitted for brevity. FDR, false discovery rate.

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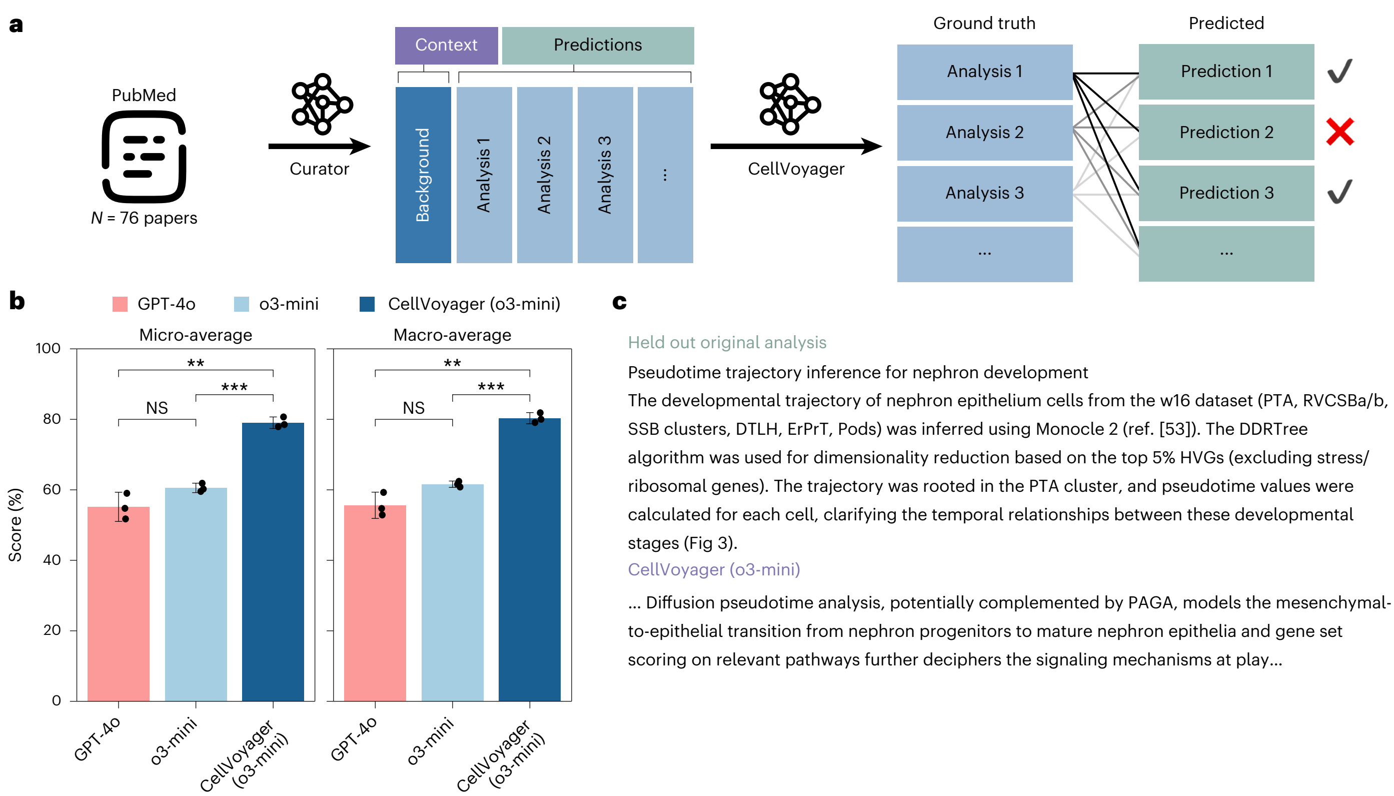

Fig. 2 | CellBench evaluation. a, Benchmark curation. CellBench is curated from 76 papers that focus on analyzing scRNA-seq data. An LLM extracts the biological background information and enumerates the independent analyses performed in the paper. The agent is tasked with predicting analyses only given the paper's background section and is evaluated on how accurately it predicts the paper's held-out analyses. **b**, Performance of base models and CellVoyager across three independent runs. With both LLMs, CellVoyager outperformed the base LLM model and CellVoyager (GPT-4o) had the highest accuracy. Bar values represent the average accuracy score in predicting a correct held-out analysis. Error bars indicate the s.d. across three independent benchmark runs. Statistical significance between methods was assessed using two-sided Welch's *t*-tests with Holm–Bonferroni correction for multiple comparisons; significance levels are denoted as follows: *$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$; NS, not significant. **c**, Example CellBench query and agent response. Here, the LLM judge labelled this as a match.

between the proposed analyses and the ground-truth analyses are measured and used as the evaluation metric. We use an LLM judge to determine whether a proposed analysis matches the ground truth. A schematic of the curation process is provided in Fig. 2a. To get a sense of how accurate the LLM judge is, two PhD students independently evaluated o3-mini's outputs on a subset of 47 generated analysis ideas; the concordances rates between each rater and the LLM judge were 89% and 85%, respectively, indicating that the LLM judge is a reliable heuristic.

Overall, CellBench is derived from 76 published studies, consisting of 659 total analyses. Given the background section of a paper, an LLM or agent is tasked with predicting the analyses conducted by the authors of the paper. As CellVoyager includes additional modules that write and correct code, for a fairer comparison with baselines, we only run the initial idea generation phase of CellVoyager, without running the deep researcher module. There, the agent finds an analysis plan and reflects upon it to find potential improvements. Additionally, we adjusted the prompt in CellVoyager to propose the most probable held-out analyses rather than solely focusing on outputting unique analyses, as the latter is biased toward more unconventional analyses. CellBench does not task the agent to write code to carry out the proposed analyses.

We evaluated CellVoyager (based on o3-mini as that is what was used in the case studies) and several base LLMs by themselves on CellBench. For all configurations, we used the default parameters from the OpenAI API endpoint and reasoning effort of medium for o3-mini. We report both the microaveraged and macroaveraged accuracies across the analyses pertaining to each paper. The results and an illustrative example are provided in Fig. 2b,c. We found that CellVoyager (o3-mini based) outperformed the GPT-4o and o3-mini base LLMs by 23.8% ($P < 0.01$) and 18.5% ($P < 0.001$) respectively. We also observed that o3-mini outperformed GPT-4o by approximately 5%, although this difference was not statistically significant. Nevertheless, given the modest performance improvement and o3-mini's strong performance on multiple coding benchmark leaderboards, we selected o3-mini for the case studies[31,32].

In total, 26 of the 76 papers were explicitly chosen to have a cutoff date at least 1 year after the training cutoff for o3-mini and GPT-4o (October 2023). We found no statistically significant difference in performance between the papers published before the training cutoff versus those published at least 1 year after the training cutoff for CellVoyager nor the base LLMs (Supplementary Table 1). Ablations of CellVoyager revealed that formulating an analysis plan and including guidelines helped drive performance (Supplementary Table 2).

Case studies

To evaluate whether CellVoyager can meaningfully extend high-quality research, we tested its ability to build upon peer-reviewed and published single-cell transcriptomics research papers. These are scenarios in which considerable effort has already been devoted toward analyzing a single-cell dataset. Therefore, it should be difficult for an agent to find interesting results related to the paper's core goals that were not already mentioned in the paper. For these case studies, we also did not give the agent access to deep research as an additional challenge. A complicating factor is that some analyses may have been conducted

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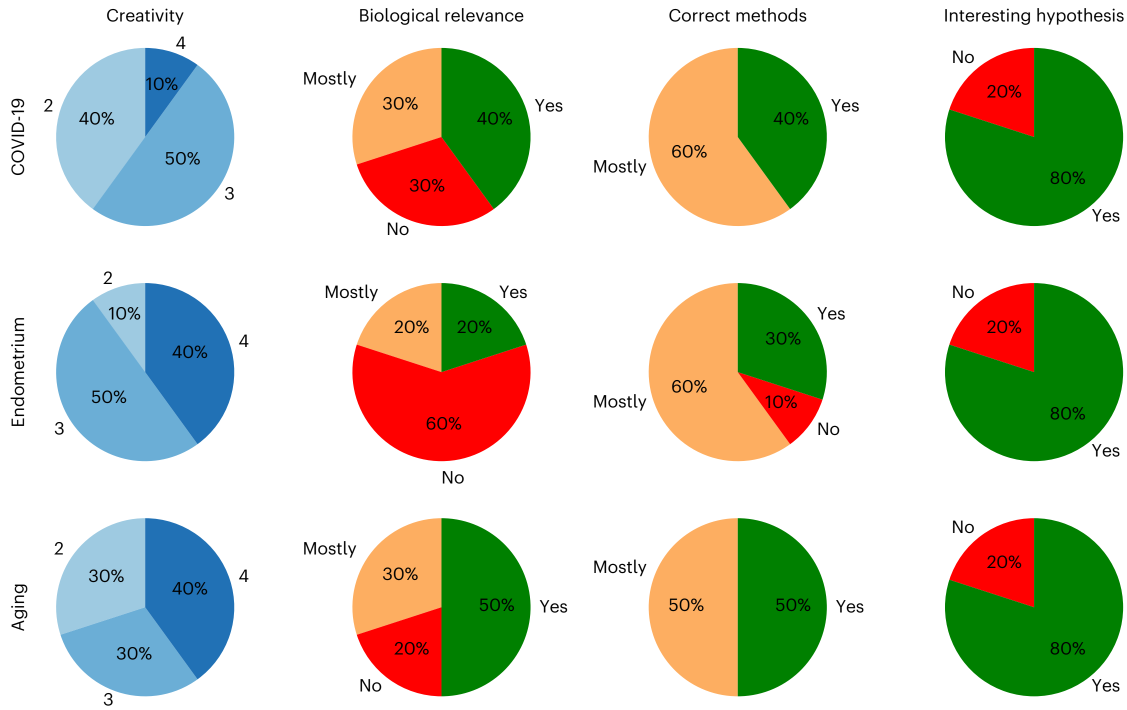

Fig. 3 | Human expert evaluations. We evaluated CellVoyager on three real-world case studies based on published papers. For each case study, the agent generated five analyses. Each case study was evaluated by two independent PhD-level researchers, one of whom was a coauthor of the original manuscript. Scores were pooled across evaluators. CellVoyager's analyses were assessed on four criteria following a standardized rubric (Supplementary Fig. 2). Section labels outside the bar correspond to the score or rating given by the evaluators. Creativity was rated on a scale of 1–4.

but were omitted from the final manuscript. To control for this, we included one of the authors of each paper in the evaluation process, ensuring familiarity with both the published analyses and any potential internal results not included in the paper.

We select three published single-cell transcriptomic studies, spanning several diseases and tissue types[4–6]. For each case study, CellVoyager generates eight distinct analyses, each encapsulated in a standalone Jupyter notebook. Each analysis code and CellVoyager's reasoning trace are documented within a standalone Jupyter notebook, typically containing 5–8 analysis steps. We selected the five analyses with the highest percentage of successfully executed code cells for further evaluation (randomly choosing among ties). We provided these Jupyter notebooks to two independent PhD-level researchers, one of whom is an author of the respective case study paper, along with an evaluation rubric (Supplementary Fig. 2). For each analysis, reviewers were asked to assess CellVoyager's creativity (on a scale of 1–4; this measures how distinct the analysis is from those in the paper), whether the analysis was biologically insightful (yes, mostly or no), whether the correct methods were used (yes, mostly or no) and whether an interesting hypothesis was raised by the agent that the evaluator would like to explore further (yes or no). To assess CellVoyager's capacity for human-guided iteration, we include feedback from the evaluators as input into CellVoyager for a subset of analyses. The agent then generated revised versions of these analyses by incorporating the reviewers' suggestions for how to improve the analyses. A summary of the evaluation results is shown in Fig. 3 with the proceeding sections going into more detail on each case study.

As a comparison, we attempted to input the same paper information and data files into Google's Data Science Agent in Colab[33] after converting them to the required formats. However, the Colab agent failed to produce meaningful results. As a general-purpose data analysis tool operating within Google Colab, it struggled significantly with single-cell data in both ideation and execution. The analyses it proposed were superficial and it had difficulty using domain-specific packages to manipulate the data effectively, often becoming trapped in repetitive error resolution loops to no avail. This observation underscores a broader limitation: generalist data science agents may lack the necessary expertise to handle complex, high-dimensional datasets such as those typical in single-cell analysis. Additionally, this highlights the value of a domain-specific single-cell agent, equipped with deeper biological context, more effective idea generation and more robust code-writing capabilities.

COVID-19 peripheral blood mononuclear cells (PBMCs). The first case study is based on a published single-cell transcriptomic atlas of PBMCs, which contains samples from seven participants with severe COVID-19 and six healthy controls[4]. The original study found depletion of multiple immune cell subsets (for example, natural killer cells and plasmacytoid dendritic cells) and the expansion of plasmablasts, particularly in participants with acute respiratory distress syndrome. Furthermore, widespread HLA-class II downregulation was observed in monocytes and B cells, while interferon (IFN)-stimulated gene signatures were seen to vary by participant age and time since the onset of the fever. Given this rich dataset and the extensive literature on COVID-19 PBMCs[34,35], there is substantial potential to explore the dataset in new ways.

CellVoyager generated five distinct analyses, each encapsulated in a standalone Jupyter notebook. These analyses were evaluated by one of the original paper's authors and by an independent PhD-level researcher using a standardized rubric (Supplementary Fig. 2). Across all analyses and evaluators, the agent achieved a mean creativity score of 2.7/4. Both evaluators independently agreed that three of the five analyses yielded biologically meaningful insights and that three analyses proposed hypotheses that were sufficiently interesting to warrant further investigation (Fig. 3).

Among the most novel analyses was one focused on pyroptosis, a highly inflammatory form of cell death, which received a perfect creativity score (4/4) and was judged to raise biologically

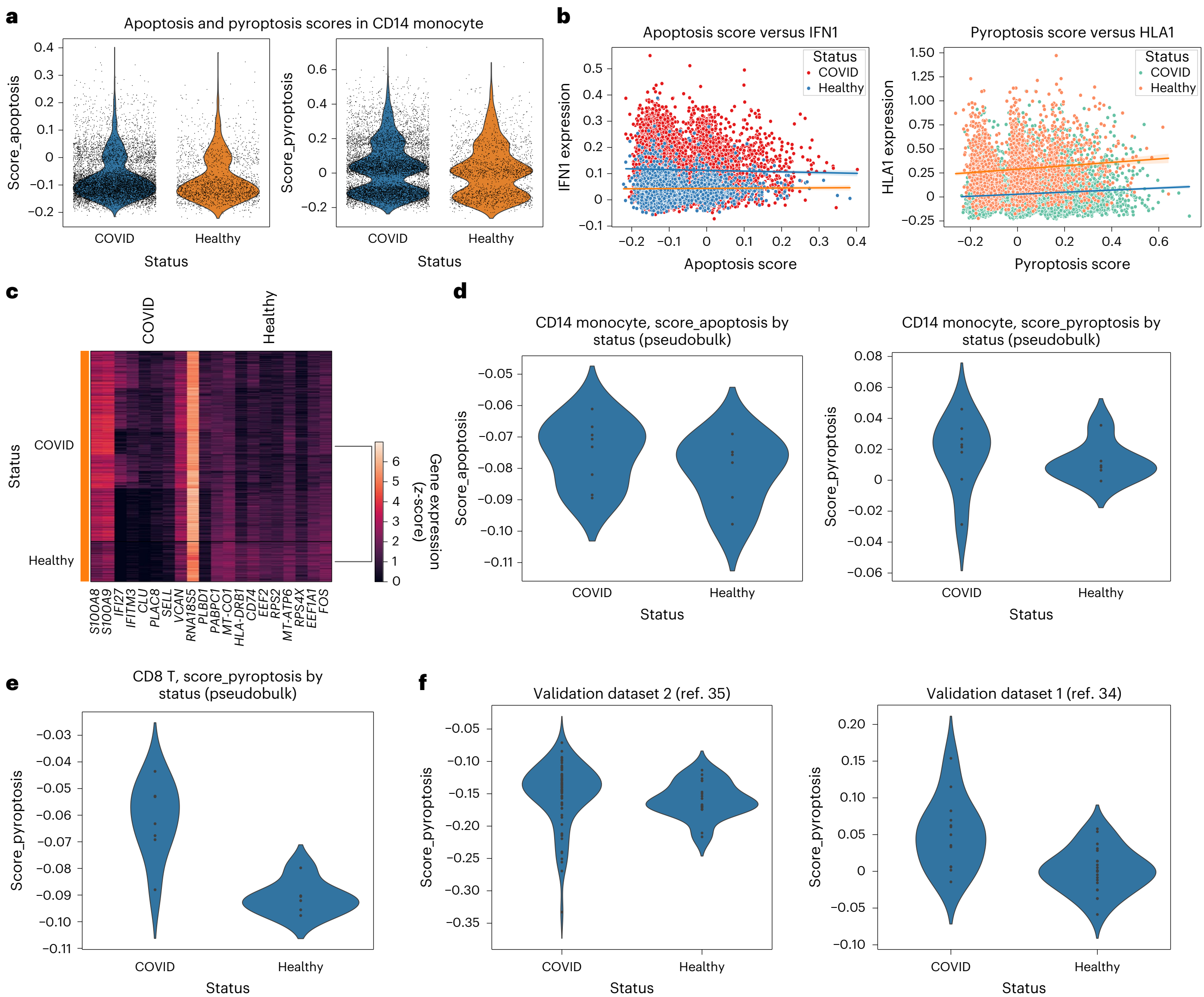

Fig. 4 | Agent-generated analysis from the COVID-19 PBMCs case study. All panels were generated by the CellVoyager agent, with some of the figure text being altered for readability. **a**, Apoptosis and pyroptosis scores. Apoptosis scores (left) and pyroptosis scores (right) are shown for $CD14^+$ monocytes in cells from participants with COVID-19 versus healthy controls. **b**, Apoptosis and pyroptosis scores versus module scores. Apoptosis versus IFN1 scores (left) and pyroptosis versus HLA1 scores (right) are shown for $CD14^+$ monocytes. Points are colored by the donor of origin, which is either a participant with COVID-19 or a healthy control. **c**, Heat map showing differentially expressed genes between high-pyroptosis scoring monocytes in COVID-19 and healthy participants. **d**, Pseudobulked comparison of apoptosis and pyroptosis scores between COVID-19 samples and healthy samples for $CD14^+$ monocytes. **e**, Pseudobulked comparison of pyroptosis scores in $CD8^+$ T cells ($P = 0.001$, according to a two-sided Mann–Whitney U-test). **f**, Analysis extended to two validation datasets where we get $P = 0.002$ and $P = 0.038$ for the left and right plots, respectively, according to a two-sided Mann–Whitney U-test.

meaningful insights (Fig. 4). Although the inputted paper did not look at pyroptosis, several subsequent publications hypothesized that SARS-CoV-2-induced pyroptosis of monocytes and that macrophages may be an important inflammatory trigger in SARS-CoV-2 infection that may drive the severity of COVID-19 disease[36–38]. Higher expression of genes involved in pyroptosis may indicate that a cell is more poised to undergo pyroptosis with the appropriate stimulation. CellVoyager began by computing apoptosis and pyroptosis scores on the basis of the expression of pathway-associated genes, comparing these between COVID-19 and healthy samples (Fig. 4a). Specifically, apoptosis scores were based on the expression of *CASP3*, *CASP8*, *BCL2*, *FADD* and *TRADD*, while pyroptosis scores were based on the expression of *GSDMD*, *CASP1*, *NLRP3* and *IL18*. CellVoyager first assessed correlations between these cell-level scores and two cell-level gene module scores computed in the original publication: a module of type I IFN-stimulated genes (IFN1) and a module of genes encoding human leukocyte antigen class II (HLA1), both of which correlated with disease severity. This analysis did not reveal a strong correlation between apoptosis or pyroptosis gene scores and IFN1 or HLA1 (Fig. 4b). The agent next isolated all high pyroptosis $CD14^+$ monocytes, split them according to whether they were from participants with COVID-19 or healthy participants and performed differential gene expression analysis between these two groups (Fig. 4c). The author of the original paper noted that the results in Fig. 4a could be more statistically rigorous by first pseudobulking each cell type by donor (Supplementary Fig. 3).

We provided this feedback to CellVoyager, which successfully completed the analysis and extended it to other cell types. There, we observed a small but insignificant increase in pyroptosis scores in

CD14⁺ monocytes from participants with COVID-19 (Fig. 4d). Interestingly, the agent found a significant increase in pyroptosis gene scores in $CD8^+$ T cells from participants with COVID-19 ($P = 0.001$) (Fig. 4e). This finding has not been prominently discussed in prior literature and may suggest that $CD8^+$ T cells in COVID-19 infection might be more poised to undergo pyroptosis. To verify this finding, we ran the CellVoyager-generated analysis on two additional independent datasets containing scRNA-seq data from PBMCs of participants with severe COVID-19 (Fig. 4f)[34,35]. In the Stephenson et al. dataset[34], we found a significant increase in the pyroptosis scores in $CD8^+$ T cells from participants with severe COVID-19 ($P = 0.002$). For the Ren et al. dataset[35], we found a smaller but still statistically significant increase ($P = 0.038$). Collectively, these results underscore CellVoyager's ability to find novel and biologically meaningful patterns in scRNA-seq data, such as the enrichment of pyroptosis signatures in $CD8^+$ T cells during severe COVID-19.

Endometrium atlas. For our second case study, we applied CellVoyager to a single-cell transcriptomics study of the human endometrium across the menstrual cycle[6]. This study used Fluidigm C1 and 10x Genomics Chromium to profile 73,180 cells from 29 healthy ovum donors sampled across the 28-day menstrual cycle. From this dataset, the original paper identified seven endometrial cell types, including a previously uncharacterized ciliated epithelium, across four major phases of endometrial transformation. Using the same procedure and evaluation protocol as in the previous case study, we sent five CellVoyager-generated analyses to two researchers, one of whom was one of the paper's coauthors, for expert review. Across the five analyses, CellVoyager received a mean creativity score of 3.3/4 and both evaluators agreed that three of the analyses presented hypotheses that were compelling and warranted further investigation (Fig. 3).

Figure 5 shows an example of an interesting new analysis conducted by CellVoyager, which was not found in the original paper. The agent begins by hypothesizing that 'paracrine signaling between stromal fibroblasts and endothelial cells … has distinct correlation patterns in early versus late phases' of the menstrual cycle. Before directly testing this hypothesis, the agent first performs several exploratory analyses. This includes visualizing the distribution of several relevant cell types across different menstrual cycle days (Fig. 5a) and visualizing stromal fibroblasts using a uniform manifold approximation and projection (UMAP), with cells colored by cluster and menstrual cycle day (Fig. 5b).

As an illustrative example, it highlights PAEP—a gene emphasized in the original study—revealing high expression in clusters such as 1 and 7 (Fig. 5c). While not intended as a novel finding, this step demonstrates the agent's ability to recognize and contextualize known phase-associated gene expression patterns. Lastly, to test its hypothesis around paracrine signaling, the agent examined coexpression patterns between receptors from endothelial cells and ligands from stromal fibroblasts. The agent plotted the coexpression of *VEGFA* versus *KDR* and *PDGFB* versus *PDGFRB*, finding positive but statistically insignificant correlations among late-phase cells (Fig. 5d). Notably, there were not enough data among early-phase cells to find early-phase correlations.

In their review of this analysis, the paper's coauthor recommended building upon the results in Fig. 5d by trying to expand the set of ligand–receptor pairs to investigate cell–cell communication (Supplementary Fig. 3). Given this feedback, CellVoyager tested 40 different ligand–receptor correlations. These consisted of five ligand–receptor pairs where the ligand came from stromal fibroblasts and the receptor was tested across eight different cell types (Supplementary Fig. 4). To facilitate interpretation, we asked the agent to extract and plot the most meaningful correlations, which are shown in Fig. 5e. TGFβ signaling between fibroblasts and other cell types across the menstrual cycle was among the top pathways identified by the agent, followed by FGF2–FGFR1 signalling across the same cell types. These pathways are recognized as key regulators of cell proliferation and tissue remodeling across multiple systems[39,40], although the role of these receptor–ligand interactions between stromal fibroblasts and smooth muscle cells, as revealed by the agent, is not fully understood. Their menstrual cycle-dependent activity is consistent with the known roles and may reflect their function in coordinating stromal–epithelial interactions during endometrial regeneration.

Aging in the brain. Our last case study builds off a paper where scRNA-seq data was generated from the subventricular zone of adult mouse brain collected across multiple ages[5]. The study identified 11 cell types, including cell types of the neural stem cell (NSC) lineage, and built single-cell aging clocks, machine learning models trained to predict age, for six of these cell types. The original dataset was used to train these aging clocks, which were then deployed to profile the effect of various rejuvenation interventions such as exercise and heterochronic parabiosis on the aging of different cell types. The original dataset was also leveraged to understand age-related gene expression trajectories and gene pathways that contributed to age prediction in the aging clocks. Given the targeted focus of the original study on building and applying aging clocks, there is substantial opportunity for new insights in reanalysis of this resource.

We again input the paper text and the processed single-cell dataset into CellVoyager and sent agent-generated analyses to the two evaluators. Across the five analyses, the agent received a mean creativity score of 3.1/4. Both evaluators independently agreed that four of the five analyses yielded some biological insights and four analyses proposed hypotheses that were sufficiently interesting to warrant further investigation (Fig. 3). An example among these was the hypothesis that 'transcriptional noise in astrocyte–qNSC cells increases with age, reflecting a loss of regulatory precision associated with cellular aging in the brain's neurogenic niche' (Fig. 6). Transcriptional noise was not analyzed in the original study but increasing transcriptional noise has been linked to aging through several recent studies[41–43]. Understanding whether such transcriptional noise also increases in astrocytes and quiescent NSCs (qNSCs) during aging is of interest in the adult mouse subventricular zone given the role that these cells have in neurogenesis, which declines with age[44].

The analysis performed by the agent begins with standard preprocessing of scRNA-seq count data, including normalization and log transformation with a pseudocount. The agent then subsets the astrocyte–qNSC cluster containing the cell type of interest and subclusters these cells further using Leiden clustering, before measuring transcriptional noise using Euclidean distance to the subcluster centroid using the top ten principal components. The approach used by the agent for measuring transcriptional noise is similar to that from previous single-cell aging studies[45–47], with a notable difference being an additional subclustering step before computing distances. The agent reported a marginal positive correlation between the transcriptional noise and age across all subclusters ($r = 0.047$, $P = 0.0135$), suggesting that there is relatively little change in transcriptional noise of astrocytes and qNSCs in the subventricular zone during aging (Fig. 6a). The observation is consistent with findings with metastudies indicating that transcriptional noise may not be a universal hallmark of aging[48]. Interestingly, the agent also discovered that the transcriptional noise varied between different subclusters of astrocytes and qNSCs, suggesting that some subpopulations may be more susceptible to these changes during aging (Fig. 6b).

Seeing little signal for astrocytes and qNSCs, the author instructed CellVoyager to extend its analysis across all cell types, prompting the agent to look at young versus old cells (split by the median age) rather than examining correlations (Supplementary Fig. 3). The agent's results in Fig. 6c demonstrate a general increase in transcriptional noise for older cells relative to younger cells across all cell types. In particular,

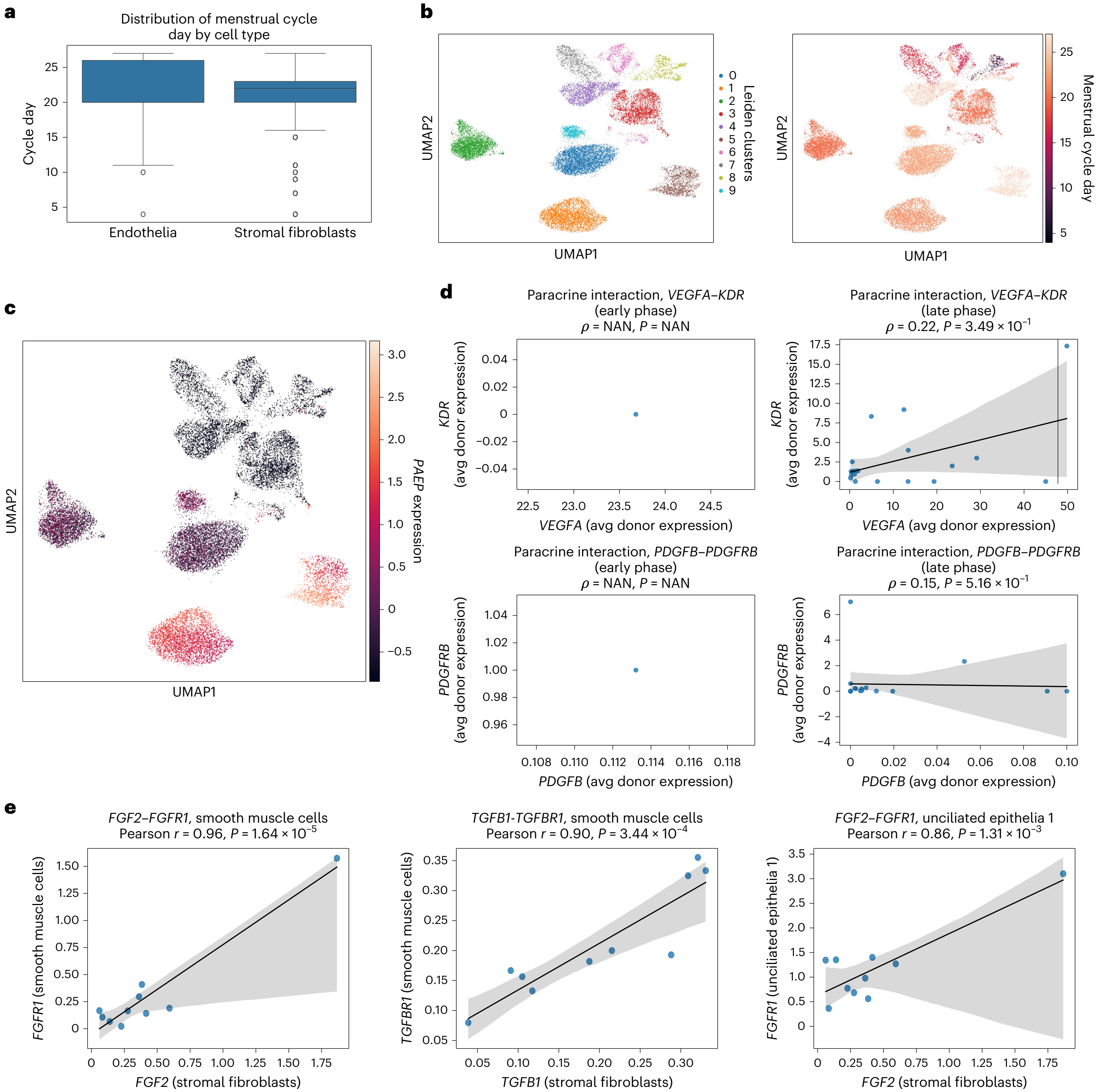

Fig. 5 | Agent-generated analysis for the human endometrium case study. All panels were generated by the CellVoyager Agent, with minor text edits applied solely for readability. **a**, Distribution of menstrual cell-cycle days for endothelia and stromal fibroblasts. Each data point corresponds to a single cell (2,060 endothelial cells and 23,063 stromal fibroblasts). The central line indicates the median, boxes denote the interquartile range (IQR) and whiskers extend to 1.5× the IQR; points beyond this range are shown as outliers. **b**, UMAP projection of the dataset colored by leiden cluster (left) and menstrual cycle day (right) **c**, UMAP plot colored by *PAEP* expression. **d**, Receptor ligand correlations between endothelial cells (receptor) and stromal fibroblasts (ligand). Each point corresponds to one donor, with expression values averaged across all cells of the indicated cell type. Top: *VEGFA* versus *KDR* correlations are shown between early-phase cells (left) and late-phase cells (right). Bottom: we have *PDGFB* versus *PDGFBR* in early-phase cells (left) and late-phase cells (right). Shaded regions indicate the 95% confidence interval of the fitted linear regression, estimated across donor-level mean expression values using bootstrapping. *P* values were computed using two-sided Pearson correlation tests. NAN, not a number. **e**, Expanded receptor ligand correlations. These are the most promising receptor–ligand correlations selected by the agent from a set of 40 receptor–ligand pairs the agent examined; the ones selected include *FGF2–FGFR1* and *TGFβ1–TGFβR1*, with the former including two different celltypes for expressing the receptor (smooth muscle cells and uncilitated epithelia). The confidence intervals and *P* values were calculated in the same manner as in **d**. avg, average.

https://doi.org/10.1038/s41592-026-03029-6

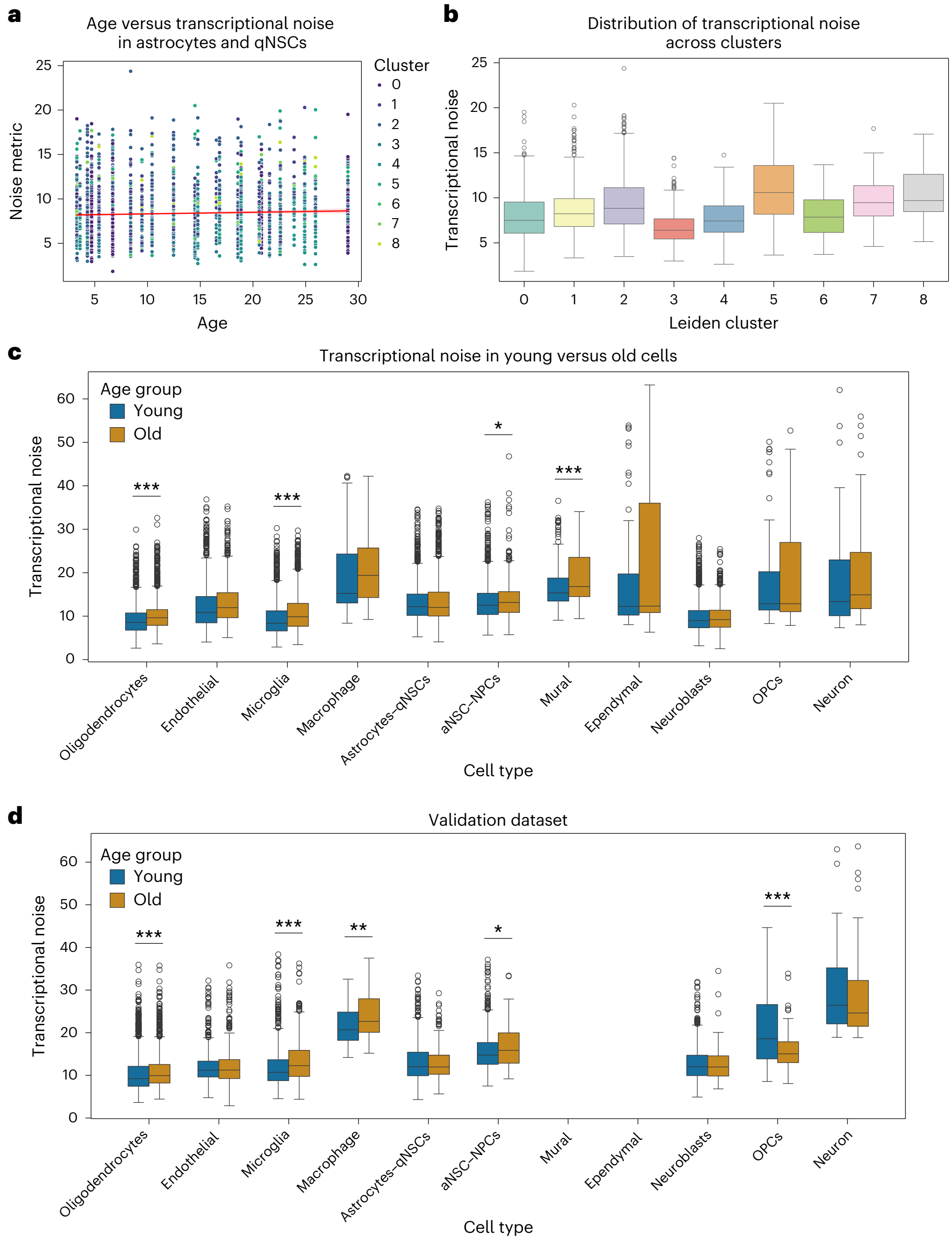

Fig. 6 | Agent generated analysis for the aging in the brain paper. All panels were generated by the CellVoyager agent, with some of the figures' text being modified for readability. For all box plots, the central line indicates the median, boxes denote the IQR and whiskers extend to 1.5× the IQR; points beyond this range are shown as outliers. **a**, Transcriptional noise versus age in the astrocyte-qNSC cluster, with points colored by their Leiden cluster. **b**, Distribution of transcriptional noise across Leiden clusters within astrocyte-qNSC cells. **c**, Transcriptional noise in young versus old cells across all celltypes in the dataset. Statistical significance was assessed by a two-sided *t*-test (*$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$). OPC, oligodendrocyte progenitor cell. **d**, Validation for transcriptional noise in young versus old cells in the Dulken et al. dataset[49], where we reused the same procedure as in the case study; cell types absent from the validation dataset are left blank.

oligodendrocytes, microglia and mural cells exhibited significantly higher transcriptional noise in older cells relative to younger cells ($P < 0.001$). Activated NSCs and neural progenitors had a smaller but still significant increase in transcriptional noise in older cells ($P < 0.05$). To validate these findings, an additional independent dataset from the subventricular zone neurogenic niche in mice was analyzed using CellVoyager's generated notebook[49]. In this dataset, all cell types that displayed significant age-associated transcriptional noise differences in the case study dataset reproduced the same significant differences, supporting the robustness of these results (Fig. 6d). Although transcriptional noise has been reported to increase with age for these cell types in multiple brain regions[50], the agent's analysis provides evidence that these age-related changes are conserved in the subventricular zone neurogenic niche and highlights the heterogeneity of transcriptional noise differences among cell types within this niche.

Discussion

CellVoyager is an LLM-based agent designed to autonomously explore scRNA-seq datasets and complement prior work conducted by human researchers. It generates new analysis ideas informed by existing biological literature and any analyses that have already been attempted. These ideas are then executed by CellVoyager, which leverages popular scRNA-seq functions and iterates within a Jupyter notebook environment. The agent synthesizes these tools to produce data-specific analyses that are biologically meaningful and extend beyond standard templates. To test CellVoyager in a challenging setting, we focus on how well CellVoyager can expand upon the results presented in a published scRNA-seq research paper, where extensive effort has been put into fully exploring the dataset. Using CellBench, we evaluated CellVoyager's ability to propose analysis ideas that are consistent with the kinds of analyses typically undertaken in single-cell studies. Our results suggest that CellVoyager can generate analysis ideas that are appropriate given a research direction, outperforming state-of-the-art LLMs in this task.

To assess CellVoyager's performance in a research setting, we conducted three case studies using published manuscripts on single-cell analysis and their associated datasets. In each case, we evaluated whether the agent could propose and execute creative, biologically meaningful analyses distinct from those included in the original papers. Authors of the original studies, all of whom hold PhDs relevant to this domain, rated each analysis using a standardized rubric. To help ensure consistency in the scoring, an additional PhD-level researcher evaluated each case study. Despite the difficulty in expanding upon a research paper, CellVoyager was able to propose creative analyses distinct from those tried in the paper (average score of 3.03/4), with, on average, 12 of the 15 total proposed hypotheses being judged to be interesting enough to warrant further investigation. CellVoyager was also able to find several new biological insights, such as higher pyroptosis scores in $CD8^+$ T cells in participants with COVID-19 relative to healthy participants, which was verified in several independent datasets. Nevertheless, the evaluators noted that several of the analyses executed by CellVoyager could be improved with a few additional steps and/or corrections. Therefore, we integrated human-in-the-loop capabilities for CellVoyager where the evaluators' comments can be given back to the agent. We demonstrated that the agent was able to meaningfully improve its analyses using just one to two pieces of feedback from the evaluators (Results. This suggests that CellVoyager benefits from some user feedback to refine its conclusions.

While CellVoyager is capable of proposing and executing meaningful analysis on single-cell data, our work had several limitations. In the current version, CellVoyager operates in a fully autonomous mode and only accepts user feedback after the analysis has completed, rather than during execution. Enabling real-time user guidance represents a natural direction for future work. Additionally, CellVoyager currently runs analyses sequentially, taking around 15–30 min to run each analysis. One possible addition to speed up run time is to identify independent lines of analysis and performing them in parallel. However, in this work, we focus more on the analytical capabilities of the agent instead of minimizing run time. Although we focus on single-cell transcriptomics, the approach generalizes to other high-dimensional biological domains. The design of CellVoyager is deliberately modular and flexible to a variety of tasks (Fig. 1), although testing on other domains was outside of the scope this study. Future work can include generalizing to other domains, such as spatial transcriptomics and proteomics. To extend CellVoyager to other data modalities, one simply needs to (1) adjust the dataset loading and summarization functions; (2) update the agent's coding guidelines to reflect the modality-specific best practices; and (3) configure which modality-specific Python packages the agent can call. In our experiments, CellVoyager often uses popular Python packages such as Scanpy. Without modification, the agent may be less able to use less popular or custom, in-house tools effectively. While this could be mitigated through fine-tuning, a potentially efficient method could be to provide tool metadata to the agent to inform its tool use[51]. A promising future direction for evaluating agents such as CellVoyager is the development of systematic replication benchmarks, where agents are tested on their ability to faithfully reproduce key results from published studies. Such benchmarks would complement discovery-focused evaluations such as those highlighted in this work. Lastly, while we designed CellVoyager to be training free and use LLMs out of the box, future work can investigate how to fine-tune modules within CellVoyager.

While CellVoyager can help users to analyze new data, our study also demonstrates that many new insights could be gained from the reanalysis of existing public data. With tens of thousands of published papers in scRNA-seq alone, thorough reanalysis by human researchers is prohibitive; however, agents such as CellVoyager could potentially unlock a treasure trove of new insights by tirelessly analyzing these vast public datasets.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-026-03029-6.

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Methods

CellVoyager implementation

Implementation of CellVoyager used Python (version 3.9.22) and the following Python packages: reportlab (version 4.4.0), nbformat (version 5.10.4) and openai (version 1.77.0). CellVoyager's own analyses used the following Python packages: scanpy (version 1.10.3), scvi-tools (version 1.1.6), celltypist (version 1.6.3), anndata (version 0.10.8), matplotlib (version 3.9.4), numpy (version 1.26.4), seaborn (version 0.13.2), pandas (version 2.2.3) and scipy (version 1.13.1). For the LLMs, we used checkpoint models, namely o3-mini-2025-01-31 and GPT-4o-2024-08-06.

An overview of the implementation of CellVoyager is highlighted in Algorithm 1 and Fig. 1. In the following sections, we go into more detail on each step of CellVoyager's procedure and showcase the prompts used for those steps.

Algorithm 1 CellVoyager analysis for a single trajectory
Require: research context C, dataset D, maximum iterations T, maximum code fix attempts F

```
π ← PLAN(C)                      ▹ Generate an exploration blueprint
NB ← INITNOTEBOOK(π, D)
for t ← 1 to T
    π ← REFLECT(π)               ▹ Self-critique plan and incorporate
                                   critiques into plan
    c ← CODE(π)                  ▹ Generate code corresponding to
                                   this step of plan
    r ← EXECUTE([NB, c])
    if r = error then            ▹ If code errored, attempt to fix
        for f = 1 to F do
            c ← FIXCODE(c, e)
            r ← EXECUTE([NB, c])
            if r ≠ error then
                break
            end if
        end for
    end if
    NB ← APPEND(NB, c′)          ▹ Add code to notebook
    π ← REPLAN(π, NB, r)         ▹ Update plan based on
                                   execution output
end for
R ← REPORT(NB)
return R, NB                     ▹ Write succinct report given notebook
```

Preprocessing. The agent loads in the metadata of the single-cell data object and summarizes it by listing column names and the first ten unique values from that column (called `adata_summary` in the prompts). Then, the agent loads in a manuscript or other text files about the biological problem or background and, optionally, a set of analyses already performed on the single-cell dataset. This information is summarized using the prompt shown in Supplementary Fig. 5 and is passed into the prompts as `paper_txt`. The names of Python packages available in a user-configured Python environment are passed into the agent. In this work, we focused on using scanpy, scvi-tools and anndata as they are popular and compatible with one another. A deep-research-style summary of general scRNA-seq analyses compatible with the packages in our environment was inputted into the agent. Lastly, the agent initializes a Jupyter notebook by loading in the single-cell dataset and importing all of the necessary packages. As the Jupyter kernel maintains state, the agent can iteratively run analyses without having to rerun previous analyses at each step.

Initial analysis plan generation. CellVoyager is designed to explore datasets as thoroughly as possible, beginning with an empty summary of prior analyses. The first idea for the analysis plan is generated using the prompt shown in Supplementary Fig. 6, which uses the system prompt shown in Supplementary Fig. 7. Optionally, the agent can run o4-mini-deep-research before idea generation, which can help with finding more biologically relevant hypotheses; if this is used, the deep research summary is appended to Supplementary Fig. 6 and all future prompts. This process yields a hypothesis, a step-by-step analysis plan, Python code for the first step, a description of that code and a brief summary (one to two sentences) of the analysis. The agent is also given a set of coding guidelines, which consists of common pitfalls observed when first designing the agent (Supplementary Fig. 8). The agent reflects on its plan by first acting as a critic of the analysis plan (Supplementary Fig. 9) and then incorporating its critiques back into the analysis plan (Supplementary Fig. 10). During this phase, the agent also inspects all single-cell-related functions used in the code cell and retrieves their documentation, a summary of which is appended to its prompt. Although the system is designed to support multiple rounds of reflection, in this work, we limited it to a single iteration to reduce runtime, as additional iterations did not yield notable improvements. Lastly, the agent adds the initial hypothesis and the full analysis plan to the Jupyter notebook.

Executing the analysis plan. The agent first adds a markdown cell to the Jupyter notebook that describes the purpose of the code for the first analysis step, followed by a code cell containing the corresponding code. It then attempts to execute the code cell. If execution fails, the agent invokes the prompt in Supplementary Fig. 11 to iteratively debug and revise the code, with a maximum of F attempts. Empirically, the success rate of code cell execution saturates after $F = 3$; thus, we by default set $F = 3$ (Supplementary Fig. 1). Upon successful execution, the textual and visual outputs of the code are passed into a vision language model using the prompt in Supplementary Fig. 12, which produces an interpretation of the results. This interpretation is appended to the notebook as a markdown cell. If the code cannot be successfully fixed after three attempts, the agent instead records a markdown cell noting that the code failed to run and that a different analytical approach should be considered.

Reevaluating the analysis plan. Following the execution and interpretation of each step in the analysis plan, the agent updates its strategy on the basis of observed results. This is achieved using the prompt shown in Supplementary Fig. 13 to generate a revised plan starting from the second step of the prior analysis (as the first step has already executed). CellVoyager then proceeds through the same iterative process described above. In this work, we limited the agent to a maximum of eight steps per analysis for efficiency. Upon completion, the agent appends the summary portion of the analysis plan to a list tracking its analysis history (inputted into the prompts as `past_analyses`), which is used as input when planning the next analyses.

CellBench

We searched for published papers that focused primarily on analyzing a scRNA-seq dataset. For 76 such papers, we used gemini-2.5-pro-preview to extract the biological background and each scRNA-seq computational analysis performed in the paper using the prompt in Supplementary Fig. 14. A subset of the curated examples were reviewed by experts to check that the extracted information accurately reflects the content of the paper.

For inference on CellBench (GPT-4o and o3-mini), we used the OpenAI API with default hyperparameters (for example, temperature of 1 for GPT-4o, reasoning effort of medium for o3-mini, etc.). We used the checkpoint models o3-mini-2025-01-31 and GPT-4o-2024-08-06 for o3-mini and GPT-4o, respectively. Inference prompts used by the base LLMs are provided in Supplementary Fig. 15. To run CellVoyager on CellBench, we adjusted the prompts to match the task of CellBench (that is, analysis prediction without any coding). In summary, identical to the typical CellVoyager workflow, the agent makes an initial plan (prompt shown in Supplementary Fig. 16), self-reflects to get feedback (Supplementary Fig. 17) and incorporates its feedback (Supplementary Fig. 18) to output its final analysis proposal.

For evaluation, we used LLM as a judge to check whether a proposed analysis is represented in the held-out set of original analyses. The LLM judge was verified by human researchers on a subset of the outputs (Results). The judge prompt is shown in Supplementary Fig. 19.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Published papers used in the case studies can be found in previous studies for COVID-19 data[4], aging data[5] and endometrium data[6]. The associated datasets can also be found in the respective publications. The CellBench evaluation dataset is available from GitHub (https://github.com/zou-group/CellVoyager).

Code availability

Code for running the agent on case studies and for CellBench evaluation is available in the CellVoyager GitHub repository (https://github.com/zou-group/CellVoyager) and at Zenodo via https://doi.org/10.5281/zenodo.17945696[52].

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Author contributions

S.A. and J.Z. conceptualized the study. S.A. implemented CellVoyager and ran it on the case studies. S.A. and B.C. developed CellBench and benchmarked CellVoyager against the base LLMs. E.S., A.I. and A.J.W. evaluated the agent's case study results. J.Z. supervised the study. All authors contributed to writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	For collecting data, we used anndata (v0.10.8), numpy (v1.26.4), pandas (v2.2.3), and scipy (v1.13.1).
Data analysis	Implementation of CellVoyager used Python (v3.9.22) the following Python packages: reportlab (v4.4.0), nbformat (v5.10.4), and openai (v1.77.0). CellVoyager's own analyses used the following Python packages: scanpy (v1.10.3), scvi-tools (v1.1.6), celltypist (v1.6.3), anndata (v0.10.8), matplotlib (v3.9.4), numpy (v1.26.4), seaborn (v0.13.2), pandas (v2.2.3), and scipy (v1.13.1).

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- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Published papers used in the case studies can be found at https://doi.org/10.1038/s41591-020-0944-y for the COVID-19 study https://doi.org/10.1038/

s43587-022-00335-4 for the aging study, and https://doi.org/10.1038/s41591-020-1040-z for the endometrium study. The associated datasets can be found in the respective publications as well. The CellBench evaluation dataset is available at https://github.com/zou-group/CellVoyager.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This information is not collected or used in this study.
Reporting on race, ethnicity, or other socially relevant groupings	This information is not collected or used in this study.
Population characteristics	This information is not collected or used in this study.
Recruitment	N/A
Ethics oversight	IRB approval was not relevant for this study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No experiments in study
Data exclusions	No experiments in study
Replication	No experiments in study
Randomization	No experiments in study
Blinding	No experiments in study

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☒	☐	Antibodies
☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☒	☐	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Plants

Seed stocks	Not used in this study
Novel plant genotypes	Not used in this study
Authentication	Not used in this study